Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	TMP
Project / Solution Name	Noka
Name (Leader)	Rohan
Email & Phone (Leader)	dr.rohansaravanan@gmail.com 0175522805
University Affiliation(s)	APU
Live Prototype URL	https://noka-ai.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/Rohansarav/noka-ai

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Noka is built as an end-to-end web platform designed to provide structured career exploration, matching, and guidance. In simple terms:

- **User Interface (Frontend):** Built with **React, TypeScript, Vite, and Tailwind CSS**. It renders the visual experience—such as the *Career Universe* and *World Dial*—allowing users to enter their profile details, view career paths, and interact with guidance features.
- **Application & API Gateway (Backend):** Powered by **FastAPI** coupled with **Pydantic v2**. FastAPI serves both the static web frontend and the API endpoints, handles private user sessions, and streams real-time updates via Server-Sent Events (SSE). Pydantic ensures all incoming and outgoing data conforms strictly to predefined schemas.
- **Core Processing & AI Services:** The backend orchestrates three specialized engines:
    1. **Deterministic Services (*Matching* + *Impact*):** Rule-based Python algorithms that align user data against standard career requirements (O*NET), scenario rules, and United Nations Sustainable Development Goals (SDG).
    2. **Tool-Using LLM Agent (*Guide*):** Uses the OpenAI SDK to extract skills, give guided step-by-step assistance, and call read-only tools within a bounded journal/execution scope.
    3. **Bounded Workflows (*Scout* + *Research*):** Gathers external, live market information by combining OpenAI models with public search tools (**Tavily** and public sources, including Malaysia-specific MyCOL data) to generate structured career proposals.
- **Database & Persistence: PostgreSQL** stores user profiles, evidence, progress logs, and execution history. The backend accesses the database through **SQLAlchemy** (ORM), while **Alembic** manages database schema updates and migrations.
- **Hosting & Deployment:** The complete unified web app and API are hosted and deployed on **Replit**.



2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11+	Backend, data processing and forecasting	PSF License
FastAPI 0.148.0	HTTP API and serving the application	MIT
Pydantic 2.14.1	Validating API and model data contracts	MIT

Name & Version	Purpose in Project	License
SQLAlchemy 2.1.5	Database access and ORM	MIT
Alembic 1.20.0	Database migrations	MIT
Psycopg 3.3.6	PostgreSQL driver	LGPL-3.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier
OpenAI API — OpenAI	Skill extraction, guided assistance and structured research/proposal generation	Build tier (top up 10& worth of credit)
Tavily Search and Extract — Tavily	Finding and extracting public web sources for career research	Free tier
SDG API — UN Statistics Division	Importing official SDG goals and targets	Public API
Indeed Hiring Lab public data	Importing historical US sector hiring index data from a published CSV	Public dataset
O*NET database downloads — U.S. Department of Labor	Importing occupation and skill reference data	Public download

3.2 Cloud Platforms & Deployment

Service & Provider	Purpose in Project	Plan / Region
Replit	Deployment	Replit core sponsored
PostgreSQL	Application database	Free version

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage	License / Terms
GPT-4.1-mini	OpenAI	Skill extraction from resumes or manually entered text	OpenAI API terms
gpt-6-sol	OpenAI	Pre-trained; Research analysis, pathway/project proposals and evidence explanations. It also used by Guide agent to select tools and retrieve application evidence	OpenAI API terms
Persistence baseline	Noka implementation	Self-trained/evaluated from historical series as a forecasting baseline **Note:** The forecasting module was initially experimented with but later removed, as its results did not provide sufficient value or meaningful improvement to the overall system's performance.	Noka implementation; no separate model licence

Model & Version	Source / Provider	Usage	License / Terms
Ridge Regression, alpha=10	scikit-learn	Self-trained by sector and forecast horizon using historical US sector data **Note:** The forecasting module was initially experimented with but later removed, as its results did not provide sufficient value or meaningful improvement to the overall system's performance.	scikit-learn BSD-3-Clause
XGBoost, 80 trees, depth 3	XGBoost	Self-trained forecasting candidate using historical US sector data **Note:** The forecasting module was initially experimented with but later removed, as its results did not provide sufficient value or meaningful improvement to the overall system's performance.	XGBoost Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type	License	Personal Data?
Indeed Hiring Lab Job Postings Tracker, US sector series	Indeed Hiring Lab public data	Public	CC BY 4.0	No

Dataset Name	Source / URL	Type	License	Personal Data?
O*NET Database 31.0; Noka-curated subset of roles and skills	O*NET Database	Public, with a project-curated subset and mappings	CC BY 4.0	No
UN Sustainable Development Goal goals and targets	UN SDG API	Public reference data	UN terms; no open licence asserted here	No
World Economic Forum Future of Jobs Report 2025	Report page	Public report used as qualitative context	Copyrighted; no open licence asserted here	No
Selected Malaysia Critical Occupations List 2024/25 facts and references	TalentCorp MyCOL publication	Publicly accessible source; selected facts/references, not a redistributed full dataset	No open licence verified	No

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use
OpenAI Codex; model varied or not recorded for all work	Code, debugging, tests and project documentation	Major
OpenAI image generation; exact model/version not recorded	Generated pitch-presentation visuals	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

I reviewed generated code and content against the product specification and available source data. The project uses schema validation for structured outputs, and career research results are presented for review rather than treated as verified facts. I also ran application and provider-flow checks. Forecasting and impact claims are limited to what the project's available data and evaluation support. I am responsible for the final submission and any corrections required.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core credits worth 20$	Deployment and hosting

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was used to deploy and host the prototype, leveraging its sponsored hosting services. While convenient for deployment, it is not essential to the core functionality, as the prototype could be hosted on alternative platforms.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Noka may store a user's self-reported skills, skill levels, career goal, target role, country or market, project details and preferences. Users may optionally submit a PDF or DOCX resume for analysis; the application extracts text and does not retain the original file bytes. When AI analysis is requested, extracted or profile context may be sent to OpenAI. Tavily receives public-search queries derived from the target role and market; resumes are not sent to search providers. Profiles are stored in PostgreSQL and associated with a server-side session identifier. The application applies ownership checks, request limits and a 30-day session expiry. Expiry restricts access; an automatic database purge schedule has not been verified.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

1. The AI can be wrong or biased
Noka's AI features can produce answers that are inaccurate, incomplete, or shaped by the data the AI learned from.
What we do: Noka checks that AI outputs follow a set structure before showing them. Research and learning plans are presented as proposals for the user to review, not as final answers.

2. Suggestions are not guarantees
A suggested learning step might look convincing, but it does not promise a job or career result.
What we do: Proposed steps are clearly framed as suggestions, and users decide what to act on.

3. Malaysia information is limited
Our Malaysia-specific content comes from public research and a few selected sources. It is not a forecast of what Malaysian employers will hire for.
What we do: We say this openly so users don't mistake it for a hiring prediction.

4. O*NET may not fit every local job
O*NET is a US database. Its job and skill mappings may not match every role in Malaysia.

What we do: Users review the results themselves, and we are upfront that the mapping is an approximation.

5. We depend on outside providers (OpenAI and Tavily)

If their service goes down, their terms change, or we hit rate limits, our AI features could be affected.

What we do: The core matching logic is deterministic and kept separate from these providers, so the main matching still works the same way even if an AI service is unavailable.

6. We haven't measured real-world impact

We have not yet measured whether Noka actually helps people find work.

What we do: We don't claim any employment benefit.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Rohan

[Signature]

[Date] 24/10/2026